

Take-Home Technical Test: Chest X-Ray Classification

Thank you for your interest in the Machine Learning Specialist role. As part of the application process, please complete the following tasks using the **Chest X-Ray Dataset** available on Kaggle:

- [Chest X-Ray Dataset](https://www.kaggle.com/datasets/muhammadrehan00/chest-xray-dataset): (<https://www.kaggle.com/datasets/muhammadrehan00/chest-xray-dataset>)
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Important Notes:

- You are welcome to use any generative AI tools and coding assistants during this assignment. However, if you advance to the next stage, you will be required and expected to demonstrate an understanding of every aspect of your submitted code
 - **We highly recommend reviewing existing machine learning works that use this dataset and building on top of them to create a deployable solution.**
 - **Only submitting notebooks (EDA, model training, etc.) will result in immediate elimination.**
 - We highly recommend following the [cookiecutter data science template](#) project structure.
 - Avoid using non-medical image augmentations that distort clinical content.
 - If you use external pretraining or models, cite sources and explain why they're appropriate.
 - Ensure runs are repeatable; set seeds and document hardware/software.
 - The work will be evaluated in three categories: Machine learning techniques, code documentation, and deployability. Each category is scored out of 10. The aggregate score is computed using the following weights: **0.35 (Machine Learning approaches)**, **0.20 (Documentation)**, and **0.45 (Deployability)**
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1) Introduction

This is an open-ended test. Demonstrate your expertise and creativity across data handling, modeling, evaluation, and deployment. You may use **any** programming language, ML framework, and tooling (e.g., PyTorch/TensorFlow, MLflow, Hydra, DVC).

2) How and What to Submit

1. A **private GitHub repository**. You can decide which files best showcase your work. That said, we recommend the following:
 - Code, configs, and documentation (README with setup & run instructions).
 - Clear **reproducibility steps** (e.g., **make train**, **make eval**) and environment files.
 - Saved metrics/artifacts, plots, and selected qualitative examples.

- *Optional*: URL to an application to demo the end-to-end mlops system

2. A short **report** (Markdown or PDF) summarizing:

- Key EDA findings and modeling choices
- Final metrics and calibration
- Robustness results (if Task 3 done)
- Explainability insights and deployment plan

3. **Access for reviewers**: Add the ML reviewers ([I-Akrout](#) and [Bgreer101](#)) as a **viewers** to your private repo.

Structure the project using a data-science cookiecutter (or a comparably clean layout) to demonstrate your ability to build reproducible codebases. .

3) Timeline

Please ensure the reviewer is added to your repository and the code is submitted by **Friday, January 16, 2026 at 11:59 PM**. **Any commits made after this time will not be considered.**

4) Selection Criteria

We will assess:

- Clarity of EDA and data handling rigor
- Soundness of modeling choices and tuning strategy
- **Metric quality, calibration, and reliability**
- Robustness and thoughtful failure analysis
- **Deployment readiness (versioning, monitoring, risk management)**
- **Code quality, reproducibility, and documentation**

5) Data Description (What to expect)

The Kaggle repository contains **chest X-ray images** organized into class-labeled folders (e.g., [NORMAL/](#) and a pathology class such as [PNEUMONIA/](#)) and split directories like [train/](#), [val/](#), [test/](#). Assume standard clinical imaging variability (different scanners, resolutions, noise) and class imbalance.

Describe, with evidence from your EDA:

- Class distribution across splits
- Image dimensions/resolutions and aspect ratios
- Metadata availability (if present) and how you'll use it
- Data quality issues (duplicates, corrupt files, extreme resizing artifacts)

Your write-up should make the dataset's statistical properties **transparent** and justify any preprocessing/curation choices.

6) Technical Tasks

Task 1 — Data Processing & EDA

1. Exploratory Data Analysis (EDA)

- Quantify class imbalance and per-split distributions.
- Visualize examples of each class and common artifacts (e.g., under/over-exposed scans).
- Summarize pixel intensity histograms and edge/texture characteristics.

2. Preprocessing Pipeline

- Standardize image sizes/aspect ratio (justify cropping vs. padding).
- Normalize intensity; consider CLAHE or similar contrast adjustments where appropriate.
- Implement train/val/test loaders with deterministic seeds, shuffling, and stratification (if you re-split).

3. Data Augmentation (Imaging-specific)

- Propose *clinically plausible* augmentations (random flips, rotations $\leq 10\text{--}15^\circ$, small translations, mild blur/noise).
- Avoid unrealistic transforms that could distort diagnostic features (e.g., aggressive warps).
- Quantify augmentation impact (e.g., ablation or training curves with/without augmentation).

4. Feature Engineering (Optional but encouraged)

- If you experiment with classical CV features (LBP/HOG, texture statistics), document rationale and results.
- Consider self-supervised embeddings (e.g., SimCLR/MoCo) pretraining on the dataset, then linear probing.

5. Missing/Corrupt Data Handling

- Define how you detect unreadable images or mislabeled samples and your remediation strategy.
- Discuss potential performance risk from label noise and what safeguards you applied.

Deliverables: Notebooks/scripts, summary figures, and a concise EDA report explaining each choice and its trade-offs.

Task 2 — Model Training & Fine-Tuning

1. Model Architecture Choice

- Start with a **transfer-learning** baseline (e.g., ResNet-50, DenseNet-121, EfficientNet-B0) initialized from ImageNet.
- Justify architecture selection for medical X-ray textures and expected data scale.

2. Training Procedure

- Loss functions (e.g., cross-entropy with class weights or focal loss for imbalance).

- Optimizer/scheduler choices and batch size vs. resolution trade-offs.
- Early stopping and callbacks (checkpointing, lr scheduling, gradient clipping).

3. Hyperparameter Tuning

- Define the search space (learning rate, weight decay, freeze depth, augmentation knobs, input resolution).
- Use a structured approach (grid/random/Bayesian); log runs with MLflow or similar.

4. Overfitting/Underfitting Controls

- Apply regularization (dropout, weight decay), differential learning rates for frozen vs. unfrozen layers, and strong validation hygiene.
- Track generalization gap; include training/validation curves and explain adjustments.

5. Evaluation Metrics & Reporting

- Report **AUC-ROC**, **F1**, **precision/recall**, **specificity/sensitivity**, **calibration** (e.g., reliability diagrams), and **confusion matrix**.
- Provide **per-class metrics** and macro/micro averages.
- Include **threshold analysis** (operating point selection for clinical risk tolerance).

Use structured experiment tracking and articulate the rationale behind each modeling decision.

(Optional) Task 3 — Holdout Set & Robustness

Create a **holdout set** different from your validation/test (e.g., images from a different hospital subset if available, or purposely altered acquisition settings).

- Evaluate performance drift and **distribution shift** (e.g., via Fréchet Inception Distance, embedding drift, or per-metric deltas).
- Diagnose performance gaps; propose mitigation (domain adaptation, stronger augmentations, test-time adaptation).

Aim to show robustness beyond the in-distribution test set.

Task 4 — Model Explainability & Safety (Imaging-specific)

- Generate **saliency**/attention maps (Grad-CAM or similar) and discuss whether highlighted regions are **plausible** given pathology.
- Document **failure modes** (false positives/negatives) with qualitative review of misclassified images.
- Describe **bias checks** (e.g., sensitivity to scanner differences, resolution, or image crop).

Explainability is crucial in medical imaging; avoid spurious cues (e.g., overreliance on text overlays or borders)

Task 5 — Deployment Strategy (Production-ready)

Propose a deployment plan for inference on new X-ray images:

- **Packaging & Versioning:** Model registry (MLflow/DVC), semantic versioning, reproducible environments.
- **Monitoring:** Data drift, performance decay, calibration tracking, and **human-in-the-loop** review for edge cases.
- **Risk & Compliance:** PHI handling, access control, audit trails, and incident response.

Focus on reliability and maintainability, not just notebook accuracy.

Good luck!